ZCAS University

School of Computing, Technology and Applied Sciences

Engineering Mathematics Final Examination (CCS 1501)

30^{th}MAY, 2025

09:00hrs - 12:00hrs

TIME ALLOWED: 3 HOURS

Instructions:

- **Section A:** This section is **Compulsory** and carries **40 Marks**.
- **Section B:** Answer any **THREE (3)** questions from this section. Each question carries **20 Marks**.
- This Examination carries a total of **100 Marks**.
- Canditates must not turn this paper until the invigilator tells them to do so.

SECTION A (Compulsory)

Question One (40 Marks)

(a) If A and B are disjoint i.e. $A \cap B = \phi$,
Simplify the following.

i. $A' \cup B'$ **[3 Marks]**

ii. $A' \cap B$ **[3 Marks]**

(b) Use the definition below to differentiate the following functions:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

i. $f(x) = x^2$ **[5 Marks]**

ii. $f(x) = 3x + 2$ **[5 Marks]**

(c) Write down the first three terms of each sequence:

i. $U_n = 3n + 5$

ii. $U_{n+1} = 5(U_n + 3), \quad U_1 = 2$ **[4 Marks]**

(d) Sketch the following piecewise function

$$f(x) = \begin{cases} 1 + x^2 & \text{if } x < 0 \\ 5 & \text{if } 0 \leq x \leq 3 \\ -1 + x & \text{if } x > 3 \end{cases}$$

[6 Marks]

(e) By using the sum to infinity formula of GPs, express each of the following in the form $\frac{a}{b}$ where a and b are relatively prime and $b \neq 0$:

i. $0.\overline{3}$ **[4 Marks]**

ii. $5.\overline{12}$ **[4 Marks]**

(f) Use the chain rule to differentiate the function $f(x) = \ln(5x^2 + 1)$ **[6 Marks]**

SECTION B (Answer any 3 questions)

Question Two (20 Marks)

(a) Convert the following from set-builder notation to roster form:

i. $C = \{z \mid z \in \mathbb{N}, z \text{ is a perfect square and } z \leq 25\}$ **[3 Marks]**

ii. $D = \{w \mid w \in \mathbb{Z}, w \text{ is odd}\}$ **[3 Marks]**

(b) The 3rd term of an AP is 1 and the 6th term is 10. Find the 4th term, 1st term and common difference. **[6 Marks]**

(c) Solve for x and y in the following:

$\frac{x}{1+i} - \frac{y}{2-i} = \frac{1-5i}{3-2i}$ **[8 Marks]**

Question Three (20 Marks)

(a) Differentiate the following using the rule:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

i. $f(x) = x^5$ **[3 Marks]**

ii. $f(x) = \frac{1}{x}$ **[3 Marks]**

(b) Express each of the following in the form $\frac{x}{y}$, where x and y are integers

(i) $0.444\ldots$ **[2 Marks]**

(ii) $0.\overline{23}$ **[2 Marks]**

(c) Discuss the function $f(x) = x^2 + 2x + 1$ **[10 Marks]**

Question Four (20 Marks)

(a) Simplify the following:

i. $\frac{2^{x+2}+2^x}{2^{x+1}}$ **[3 Marks]**

ii. $\frac{18^x-9^x}{3^x}$ **[3 Marks]**

(b) Solve for x:

i. $\log(x^2-4) = \log(x+2)$ **[4 Marks]**

ii. $(x+iy)(4i) = 5$ **[3 Marks]**

(c) Use set notation to describe the set of values of x for which: $x^2+2x-3<0$ and $x-4>1-2x$. **[7 Marks]**

Question Five (20 Marks)

(a) Evaluate without using a calculator:

i. $\log_5 125 - \log_2 16 + \ln e^3$ **[4 Marks]**

ii. $-\ln e^2 + 2\ln e + 1$ **[4 Marks]**

(b) Given that X and Y are any two sets, simplify the following:

i. $X' \cap (X \cap Y')'$ **[3 Marks]**

ii. $[(X \cap Y) \cup (X \cap Y')]'$ **[4 Marks]**

(c) Determine the coefficient of x^3 in the binomial expansion of $(2+3x)^6$. **[5 Marks]**

End...

This too shall pass!!!

Best wishes!

@2025